LineWatch

Recipe-aware inspection at the edge

Rockwell Automation × NVIDIA Physical AI Hackathon · MSOE

Reagan Burkemper · Cameron Delpage · Hayden Fons · Zoe Kirkman · Srijan Kumar · Josiah Mathews · Alayna Ostler · Austin Koske

LineWatch catches off-recipe vials while they are being filled. Using the station's existing cameras and the recipe order, it tells the operator to scrap, flush or hold, with a deadline and a reason. It is trained only in simulation and runs on an NVIDIA GB10.

76% → 94%
out-of-spec vials caught
color check vs LineWatch

8 s
fill end to the call
median; 120 s deadline, 100+ s spare

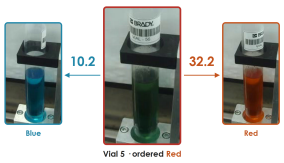
0 s
added per vial
vs 20+ s at CSI's sorter

$10M
average food recall
caught at the fill instead

Same 202 held-out synthetic out-of-spec vials for both catch rates. Decision time: Run 5, every SCRAP before its vial reached the tray. Recall cost: FMI/GMA.

THE MANUFACTURING PROBLEM

CSI's Fluid Mixing Station 3 fills each vial by recipe from four dye tanks (clear, orange, yellow, blue; yellow was down for our tests); a conveyor and robot arm then carry it to a sorter that drops it in a Good or Bad tray. Lines that run several recipes on shared equipment leave old product in the line at every changeover: **carryover**. CSI's sorter is never told the order, so a vial of the previous recipe still passes as a valid color, and it stops every vial for over 20 s to say only pass or fail.



Vial 5 at CSI: red ordered, blue still in the line.

TECHNICAL APPROACH

1. **Perception.** A YOLO11n detector (COCO-pretrained) finds the vial; a temporal U-Net segments the liquid across consecutive frames as it fills. Both are trained only on Isaac Sim renders.
2. **Measurement QA.** Glare, clipping, too few liquid pixels or an unstable track give *can't verify*, never a guess.
3. **Spec check.** The fill-end color is compared with one reference vial of the ordered recipe: more than ΔE_{00} 5 away is out of spec (ΔE_{00} is the standard color-difference measure; 5 is clearly visible side by side).
4. **Carryover.** If the shift points toward the previous recipe (cosine of at least 0.75 in Lab or optical density, where dyes mix linearly), the cause is carryover.
5. **Action.** SCRAP before the sorter (120 s deadline), FLUSH before the next fill, STOP after 3 deviations, or HOLD. Nemotron words the explanation; the rules decide. Our rule explainer matched the evidence on 20/20 vials, Cosmos-Reason1 on 6–9/20.

HOW WE USED THE CSI EQUIPMENT

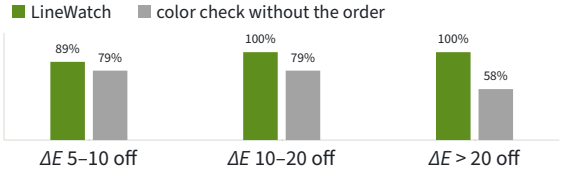
- Ran on **CSI's GB10** with the station's **fill and tray cameras**: 3 visits (5 h) of integration testing, then final runs (4 blue vials, then 4 red).

HOW WE USE PHYSICAL AI

- **Sim-to-real:** trained on Isaac Sim renders of the station's twin, run on CSI's real cameras.
- **Synthetic data:** 14,400 carryover renders on NVIDIA Brev set the thresholds.
- **Edge AI:** 86 ms per frame on the GB10; 0 of 50,941 frames dropped over Run 5 at real-time speed.
- **Bring your own line:** new recipe, one reference vial (no retraining); new station, a config file and a short tray video; new product, render its twin and retrain.

EVIDENCE THAT IT WORKS

Out-of-spec vials caught on 1,195 held-out changeovers



The color check gets *worse* on big shifts: they look like another valid recipe.

	Synthetic held-out	**Run 5 +** carryover added	**Run 5** as recorded	**CSI** final run
Out-of-spec caught	189/202	5/5	7/11	**4/4**
False scraps	4.3% (43/993)	1/16	0/9	**0/4**
Baseline caught	154/202	5/5	6/11	4/4
SCRAP before the tray	–	–	9/9	**4/4**

Run 5 is the organizers' 30-vial run; *carryover added* recolors its real footage with known amounts. Real vials were judged by our team with the measured ΔE. Baseline: passes any vial within ΔE 5 of a known recipe, without the order.

HOW WE SCORED IT

Truth is the spec: a vial is out of spec when its true color, the optical-density mix of the clean and previous recipes at the true residue, is more than ΔE_{00} 5 from the order. Measured colors get camera noise matched to Runs 1–4. Thresholds were chosen on training scene families and reported once on held-out ones; Run 5 was scored once with settings frozen.
Limits: misses sit within camera noise of the line; vials with too little visible liquid get no call; sorting needs PLC access.

Code GitHub repository **Results** evaluation.md **Design** architecture.md **Data** data.md **Sources** industry-research.md **Video** youtu.be/byBT2FvAK1E